

Two-Mode Active-Subspace Reduction and Reciprocal Determinant Thresholds in Linear Ornstein–Uhlenbeck Systems

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First draft — September 2026

Abstract

We study the stationary covariance of a stable linear Ornstein–Uhlenbeck system whose drift, restricted to a two-dimensional invariant subspace, has distinct relaxation rates $D_1, D_2 > 0$, and whose forcing on that subspace is a rank-one deformation of an isotropic baseline, $Q = q_b(I + (\tau - 1)ee^\top)$, along a unit *excitation direction* $e = cu + sv$. Here $\tau > 0$ is a *variance* ratio. Solving the Lyapunov equation on the plane gives an exact 2×2 block, and we prove that its determinant is an exactly *palindromic* quadratic in τ :

$$4D_1D_2 \det \Sigma = q_b^2(\mathcal{A}\tau^2 + (1 - 2\mathcal{A})\tau + \mathcal{A}), \quad \mathcal{A} = c^2s^2\nu^2, \quad \nu = \frac{D_1 - D_2}{D_1 + D_2}.$$

The leading and constant coefficients coincide because the forcing family is self-dual, $Q_e(\tau) = \tau Q_{e^\perp}(1/\tau)$, which is $ee^\top + e^\perp e^{\perp\top} = I$ in disguise. Only then does the reciprocal coordinate $w = \tau + 1/\tau$ appear: dividing by τ makes the normalized determinant exactly affine in w , which yields an exact generalized-variance threshold. The coefficient $\mathcal{A}$ vanishes precisely when the excitation lies in a single drift eigenmode or the two rates coincide, and is maximized at equal modal weights. Graph-Laplacian systems give corollaries: the two-node model of earlier work is recovered with $\tau = r^2$, and P_3 and K_3 give exact optimal thresholds $35 + 15\sqrt{5}$ and $247/2$. The central identities are machine-checked in Lean 4 with mathlib. We emphasize that the one-coordinate closure is tied to an effectively two-dimensional active structure and does not extend automatically to genuinely multi-mode excitation.

1 Introduction

Linear Ornstein–Uhlenbeck (OU) systems are among the few multivariate stochastic models whose stationary statistics are available in closed form: for a Hurwitz drift the stationary covariance solves an algebraic Lyapunov equation [19, 11, 20]. This makes them a natural setting in which to ask how *heterogeneous* forcing interacts with coupling, and the question has been studied directly: Barucca analyses localization in covariance matrices of coupled heterogeneous OU processes [3]; Ferreira, Metz and Barucca develop a random-matrix ensemble for OU covariances with heterogeneous temperatures [9]; Godrèche and Luck give exact nonequilibrium stationary states for linear Langevin systems at two temperatures [12]; and the Brownian-gyrator literature studies the same two-temperature structure in a planar model [10, 8].

A scalar summary of such a covariance is its determinant, Wilks’s generalized variance [22], which is monotone in the differential entropy of the stationary Gaussian [5]. Earlier work of the author [7]

analysed two diffusively coupled OU processes with unequal noise amplitudes and found that this determinant can exceed its uncoupled value at finite coupling if and only if the noise-amplitude ratio exceeds $3 + 2\sqrt{2}$ — an exact algebraic threshold in a two-node model.

The starting point of the present paper is that the two-node graph is *not* the essential structure. What matters is a geometric configuration that a two-node system happens to realize: a two-dimensional invariant subspace of the drift on which the relaxation rates differ, carrying a forcing that is isotropic except for a rank-one deformation along one direction. We call that subspace the *active plane* and that direction the *excitation direction*. Under those hypotheses — and with no reference to any graph — the stationary covariance determinant is exactly palindromic in the variance ratio, and therefore collapses onto the single reciprocal coordinate $w = \tau + 1/\tau$.

The logical order matters and we keep it throughout. The reciprocal coordinate is not posited as a convenient change of variable: it is *forced* by an algebraic symmetry of the determinant, which is in turn forced by a self-duality of the rank-one forcing family. The chain is

$$\begin{aligned} \text{rank-one excitation} &\rightarrow \text{two-mode covariance block} \rightarrow \text{palindromic determinant} \\ &\rightarrow w = \tau + 1/\tau \rightarrow \text{exact threshold,} \end{aligned}$$

and each arrow is proved before the next is used.

What is standard and what is not. The ingredients are almost entirely classical: the OU stationary covariance and the Lyapunov equation [11]; the modal solution $\Sigma_{ij} = Q_{ij}/(D_i + D_j)$, standard in the heterogeneous-OU literature [3, 12]; the reduction of a self-reciprocal polynomial through $t + 1/t$ [21]; the trace/determinant shape statistic and its relation to Mauchly’s sphericity test [14, 15]; and the Newton-identity obstruction in dimension $n \geq 3$. What we did not locate in the literature we searched is the specific assembly: the exact palindromic determinant identity, its reciprocal reduction, and the resulting exact generalized-variance threshold optimized over a coupling parameter. We state that conservatively in Section 13 and claim nothing stronger.

Scope. All results are exact real algebra about a linear-Gaussian model. Nothing here is asserted about nonlinear, biological, cognitive or social systems, and no physical universality is attributed to any of the constants that appear.

2 Setting and notation

Consider a linear stochastic differential equation on $\mathbb{R}^n$,

$$dX = MX dt + B dW, \quad Q := BB^\top, \quad (1)$$

with M Hurwitz. The stationary covariance Σ is the unique solution of the algebraic Lyapunov equation

$$M\Sigma + \Sigma M^\top = -Q. \quad (2)$$

Existence and uniqueness of the stationary Gaussian law for (1) is classical [11, 20]; we take (2) as the object of study and prove statements about it.

Definition 1 (Active plane and excitation direction). Let $\Pi \subset \mathbb{R}^n$ be a two-dimensional subspace invariant under M , on which M acts symmetrically with eigenmodes u, v and relaxation rates $D_1, D_2 > 0$; that is, in the orthonormal basis (u, v) of Π ,

$$M|_\Pi = \text{diag}(-D_1, -D_2).$$

Active plane: drift eigenbasis and rank-one excitation

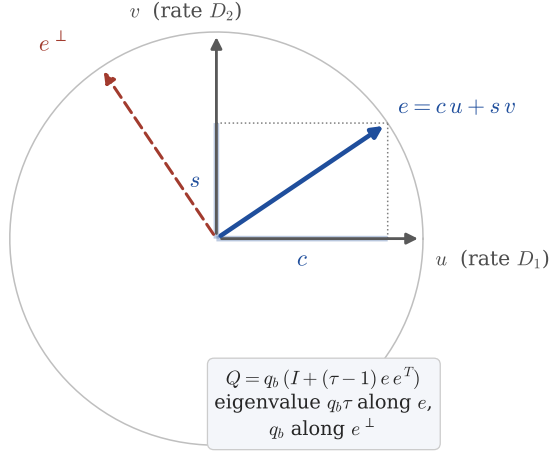


Figure 1: The active plane. The drift is diagonal in the eigenbasis (u, v) with rates D_1, D_2 ; the forcing is isotropic except for a rank-one deformation of relative size τ along the excitation direction $e = cu + sv$. The mixing $c^2 s^2$ and the rate contrast ν are the only features of this picture that enter the determinant.

Let $e = cu + sv$ with $c^2 + s^2 = 1$ be a unit vector in Π , the *excitation direction*, and let $e^\perp = -su + cv$. We call Π the *active plane*. Throughout, e denotes this direction and never Euler's number.

Definition 2 (Rank-one heterogeneous forcing). For a baseline intensity $q_b > 0$ and a parameter $\tau > 0$, the forcing on Π is

$$Q(\tau) = q_b(I + (\tau - 1)ee^\top), \quad (3)$$

whose eigenvalues on Π are $q_b\tau$ along e and q_b along $e^\perp$.

Remark 3 (τ is a variance ratio). This point is a frequent source of confusion and we fix it now. The parameter τ in (3) is a ratio of *forcing eigenvalues*, i.e. of noise *variances* (or temperatures), not of noise *amplitudes*. When the two-node model of [7] is recovered in Section 11.1, its amplitude ratio r satisfies

$$\tau = r^2, \quad \text{hence} \quad \tau + \frac{1}{\tau} = \left(r + \frac{1}{r}\right)^2 - 2. \quad (4)$$

So $w = \tau + 1/\tau$ and $s(r) = r + 1/r$ are *different* reciprocal invariants of the same physical situation, related by (4); conflating them misreports a threshold by a square. We always write w for the variance invariant. This remark fixes notation only: the coordinate w plays no role until Section 6, where it is *forced* by the palindromic identity rather than chosen for convenience.

We write ν for the rate contrast and $\mathcal{A}$ for the coefficient that will govern everything:

$$\nu = \frac{D_1 - D_2}{D_1 + D_2}, \quad \mathcal{A} = c^2 s^2 \nu^2. \quad (5)$$

Figure 1 shows the configuration.

3 The active covariance block

Theorem 4 (Stationary covariance on the active plane). *Let $D_1, D_2 > 0$ and let Q be any symmetric forcing on Π with entries q_{11}, q_{12}, q_{22} in the basis (u, v) . Then (2) restricted to Π has the unique solution*

$$\Sigma = \begin{pmatrix} \frac{q_{11}}{2D_1} & \frac{q_{12}}{D_1 + D_2} \\ \frac{q_{12}}{D_1 + D_2} & \frac{q_{22}}{2D_2} \end{pmatrix}, \quad \text{that is,} \quad \Sigma_{ij} = \frac{Q_{ij}}{D_i + D_j}. \quad (6)$$

For the rank-one forcing (3) this reads

$$\Sigma = \begin{pmatrix} \frac{q_b(1 + (\tau - 1)c^2)}{2D_1} & \frac{q_b(\tau - 1)cs}{D_1 + D_2} \\ \frac{q_b(\tau - 1)cs}{D_1 + D_2} & \frac{q_b(1 + (\tau - 1)s^2)}{2D_2} \end{pmatrix}. \quad (7)$$

Proof. With $M|_\Pi = \text{diag}(-D_1, -D_2)$, equation (2) reads entrywise $-(D_i + D_j)\Sigma_{ij} = -Q_{ij}$, which determines each Σ_{ij} uniquely because $D_i + D_j > 0$. Substituting the entries of (3) gives (7). $\square$

Equation (6) is the standard modal form of the Lyapunov solution for a symmetric drift and is used throughout the heterogeneous-OU literature [3, 12, 9]; we claim no novelty for it. It is recorded as a theorem only because everything below is computed from it.

Lemma 5 (Forcing determinant). *For $c^2 + s^2 = 1$, $\det Q(\tau) = q_b^2 \tau$.*

Proof. $Q(\tau)$ has eigenvalues $q_b \tau$ and q_b ; equivalently, expanding the 2×2 determinant of (3) and using $c^2 + s^2 = 1$ gives $q_b^2 [(1 + (\tau - 1)c^2)(1 + (\tau - 1)s^2) - (\tau - 1)^2 c^2 s^2] = q_b^2 \tau$. $\square$

4 Self-duality of the forcing family

The palindromic structure of the next section is not an accident of the algebra; it descends from a symmetry of the family (3) itself.

Proposition 6 (Self-duality). *Let $c^2 + s^2 = 1$ and $\tau > 0$. Then*

$$Q_e(\tau) = \tau Q_{e^\perp} \left(\frac{1}{\tau} \right), \quad (8)$$

where $Q_{e^\perp}(\sigma) = q_b(I + (\sigma - 1)e^\perp e^{\perp\top})$.

Proof. Because $(e, e^\perp)$ is an orthonormal basis of Π ,

$$ee^\top + e^\perp e^{\perp\top} = I. \quad (9)$$

Hence $q_b(I + (\tau - 1)ee^\top) = q_b(\tau ee^\top + e^\perp e^{\perp\top})$, and likewise $\tau q_b(I + (\tau^{-1} - 1)e^\perp e^{\perp\top}) = \tau q_b(ee^\top + \tau^{-1}e^\perp e^{\perp\top}) = q_b(\tau ee^\top + e^\perp e^{\perp\top})$. The two agree. $\square$

Equation (8) says that inverting the heterogeneity ratio and rotating the excitation by a quarter turn returns the same forcing up to an overall scale τ . Since the determinant of a 2×2 matrix scales as τ^2 under $Q \mapsto \tau Q$, and since Σ depends linearly on Q by Theorem 4, the map $\tau \mapsto 1/\tau$ can only act on $\det \Sigma$ by a factor τ^2 together with the exchange $c \leftrightarrow s$. The coefficient $\mathcal{A} = c^2 s^2 \nu^2$ is symmetric in $c \leftrightarrow s$, which is precisely why the exchange leaves the polynomial invariant and makes it palindromic.

Remark 7. Self-duality of this kind is elementary and we do not present it as a new concept; identity (9) is the resolution of the identity on a plane. The contribution here is its role as the structural cause of the determinant identity in Section 5.

5 The palindromic determinant

Theorem 8 (Palindromic determinant identity). *Let $D_1, D_2 > 0$, $c^2 + s^2 = 1$, $q_b \in \mathbb{R}$, $\tau \in \mathbb{R}$, and let Σ be as in (7). Then*

$$4D_1D_2 \det \Sigma = q_b^2 \left(\mathcal{A} \tau^2 + (1 - 2\mathcal{A}) \tau + \mathcal{A} \right), \quad \mathcal{A} = c^2 s^2 \nu^2. \quad (10)$$

In particular the quadratic in τ is palindromic: its leading and constant coefficients are equal.

Proof. Write $\Delta = D_1 + D_2$. From (7),

$$\det \Sigma = \frac{q_b^2 (1 + (\tau - 1)c^2)(1 + (\tau - 1)s^2)}{4D_1D_2} - \frac{q_b^2 (\tau - 1)^2 c^2 s^2}{\Delta^2}.$$

Multiplying by $4D_1D_2\Delta^2$ and using Lemma 5 in the form $q_b^2(1 + (\tau - 1)c^2)(1 + (\tau - 1)s^2) - q_b^2(\tau - 1)^2 c^2 s^2 = q_b^2 \tau$ gives

$$\begin{aligned} 4D_1D_2\Delta^2 \det \Sigma &= \Delta^2 \left[q_b^2 \tau + q_b^2 (\tau - 1)^2 c^2 s^2 \right] - 4D_1D_2 q_b^2 (\tau - 1)^2 c^2 s^2 \\ &= q_b^2 \left[\Delta^2 \tau + c^2 s^2 (\tau - 1)^2 (\Delta^2 - 4D_1D_2) \right] \\ &= q_b^2 \left[\Delta^2 \tau + c^2 s^2 (D_1 - D_2)^2 (\tau - 1)^2 \right], \end{aligned} \quad (11)$$

since $\Delta^2 - 4D_1D_2 = (D_1 - D_2)^2$. Dividing by Δ^2 and using $\mathcal{A} = c^2 s^2 (D_1 - D_2)^2 / \Delta^2$,

$$4D_1D_2 \det \Sigma = q_b^2 [\tau + \mathcal{A}(\tau - 1)^2] = q_b^2 [\mathcal{A} \tau^2 + (1 - 2\mathcal{A}) \tau + \mathcal{A}]. \quad \square$$

Two features of (11) deserve comment. First, the entire dependence on the rates enters through the single combination $(D_1 - D_2)^2 / (D_1 + D_2)^2 = \nu^2$: the determinant does not see D_1 and D_2 separately beyond the overall factor $4D_1D_2$. Second, the heterogeneity enters only through $(\tau - 1)^2$, which is manifestly symmetric under the substitution that sends $\tau - 1$ to $\tau(\tau^{-1} - 1)$; this is the self-duality of Proposition 6 reappearing as the palindromic symmetry of the coefficients.

6 The reciprocal reduction

Only now, with palindromicity established, do we introduce the reciprocal coordinate. A palindromic polynomial of even degree $2m$ in t is a polynomial of degree m in $t + 1/t$ after division by t^m ; for $m = 1$ this is immediate, and the general statement is classical (see [21, §2], where the substitution $x = z + 1/z$ and its Chebyshev connection are set out). We use only the $m = 1$ case, which we prove in line; the citation is for provenance of the substitution, not for the identity below.

Definition 9. For $\tau > 0$ set $w = \tau + \frac{1}{\tau}$.

Lemma 10. *For $\tau > 0$, $w - 2 = (\tau - 1)^2 / \tau \geq 0$, with equality if and only if $\tau = 1$. Hence $w \geq 2$ always, and $w = 2$ exactly in the homogeneous case.*

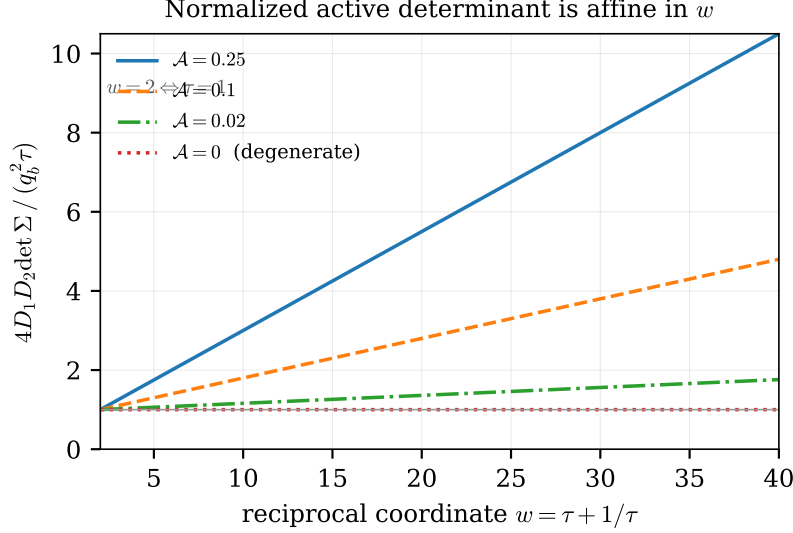


Figure 2: The normalized active determinant (12) as a function of the reciprocal coordinate w , for several values of $\mathcal{A}$. The dependence is exactly affine, with slope $\mathcal{A} \in [0, 1/4)$ and value 1 at $w = 2$ (the homogeneous point $\tau = 1$).

Theorem 11 (Reciprocal reduction). *Let $D_1, D_2 > 0$, $c^2 + s^2 = 1$, $q_b \neq 0$, $\tau > 0$. Then*

$$\frac{4D_1 D_2 \det \Sigma}{q_b^2 \tau} = 1 + \mathcal{A}(w - 2). \quad (12)$$

Proof. Divide (10) by $q_b^2 \tau$, which is nonzero:

$$\frac{\mathcal{A}\tau^2 + (1 - 2\mathcal{A})\tau + \mathcal{A}}{\tau} = \mathcal{A}\tau + 1 - 2\mathcal{A} + \frac{\mathcal{A}}{\tau} = 1 + \mathcal{A}\left(\tau + \frac{1}{\tau} - 2\right). \quad \square$$

The normalization in (12) is natural: $q_b^2 \tau = \det Q$ by Lemma 5, and $4D_1 D_2 = \prod_{i,j} (D_i + D_j)^{1/2} \dots$ — more simply, $q_b^2 \tau / (4D_1 D_2)$ is exactly $\det \Sigma$ in the homogeneous-direction case $\mathcal{A} = 0$. So (12) compares the determinant against what it would be with no reciprocal contribution, and the comparison is *exactly affine* in w (Figure 2).

7 Degeneracy and size of the coefficient

Everything in (12) is controlled by the single number $\mathcal{A} = c^2 s^2 \nu^2$. We record exactly when it vanishes and how large it can be.

Proposition 12 (Exact degeneracy conditions). *Let $D_1, D_2 > 0$ and $c^2 + s^2 = 1$. Then $\mathcal{A} \geq 0$, and*

$$\mathcal{A} = 0 \iff c = 0 \text{ or } s = 0 \text{ or } D_1 = D_2. \quad (13)$$

Separately, the reciprocal contribution $\mathcal{A}(w - 2)$ vanishes if and only if $\mathcal{A} = 0$ or $\tau = 1$.

Proof. $\mathcal{A} = c^2 s^2 \nu^2$ is a product of squares, hence nonnegative, and vanishes iff $c = 0$, $s = 0$ or $D_1 - D_2 = 0$ (note $D_1 + D_2 > 0$, so ν is well defined and $\nu = 0$ iff $D_1 = D_2$). The second statement follows from Lemma 10. $\square$

The three degeneracies have clean readings, and together they say exactly what the mechanism requires:

- $c = 0$ or $s = 0$: the excitation lies in a *single* drift eigenmode. The forcing is then simultaneously diagonalizable with the drift, the covariance is diagonal, and heterogeneity cannot couple the modes.
- $D_1 = D_2$: no relaxation contrast. The active block is a multiple of the identity in its rate structure, and the determinant is blind to how the forcing is oriented.
- $\tau = 1$: no forcing heterogeneity at all.

Thus a nonzero reciprocal contribution requires *both* genuinely mixed excitation and genuinely unequal rates. This is the sharpest explanatory content of the theorem: the effect is an interference between two anisotropies — one in the drift, one in the forcing — and it disappears if either is absent.

Proposition 13 (Modal-mixture bound). *If $c^2 + s^2 = 1$ then $c^2 s^2 \leq \frac{1}{4}$, with equality if and only if $c^2 = s^2 = \frac{1}{2}$. If moreover $D_1, D_2 > 0$ then $\nu^2 < 1$. Consequently*

$$0 \leq \mathcal{A} < \frac{1}{4}. \quad (14)$$

Proof. $\frac{1}{4} - c^2 s^2 = \frac{1}{4} - c^2(1 - c^2) = (c^2 - \frac{1}{2})^2 \geq 0$, with equality iff $c^2 = \frac{1}{2}$. For the rate contrast, $(D_1 + D_2)^2 - (D_1 - D_2)^2 = 4D_1 D_2 > 0$, so $\nu^2 < 1$ strictly. $\square$

Remark 14. The bound $c^2 s^2 \leq 1/4$ says the heterogeneity-sensitive coefficient is largest when the excitation projects equally onto the two active modes. We read this as a statement about this determinant and nothing more; it is not proposed as a physical optimality principle. Note also that $\mathcal{A} < 1/4$ is strict: $\nu^2 < 1$ for any finite positive rates, and $\nu^2 \rightarrow 1$ only in the singular limit $D_2/D_1 \rightarrow 0$ or ∞ .

8 The threshold theorem

In applications the active block sits inside a larger system and one compares the determinant at coupling κ against a reference. Both effects — the shrinking of the homogeneous determinant and the reciprocal enhancement — combine into a single scalar comparison.

Theorem 15 (Exact threshold). *Let $\Phi > 0$ and $C > 0$ and consider*

$$V = \Phi(1 + C(w - 2)). \quad (15)$$

Then

$$V > 1 \iff \Phi C(w - 2) > 1 - \Phi \iff w > w^*, \quad w^* = 2 + \frac{1 - \Phi}{\Phi C}. \quad (16)$$

Proof. The first equivalence is a rearrangement of (15). For the second, the identity

$$w - \left(2 + \frac{1 - \Phi}{\Phi C}\right) = \frac{\Phi(1 + C(w - 2)) - 1}{\Phi C}$$

holds identically, and $\Phi C > 0$, so the left side is positive exactly when the numerator on the right is. $\square$

Remark 16 (Edge cases, stated rather than hidden). Three cases deserve explicit mention.

1. If $\Phi \geq 1$ then $w^* \leq 2$. Since $w \geq 2$ always (Lemma 10), the threshold is then vacuous: the determinant already exceeds the reference at $w = 2$, with no heterogeneity required. In the graph applications below $\Phi < 1$ for $\kappa > 0$, so this does not occur there, but the general statement must allow it.
2. If $C = 0$ the equivalence (16) is unavailable — (15) reduces to $V = \Phi$, independent of w , and no amount of heterogeneity helps. This is exactly the degenerate case of Proposition 12.
3. w^* is a threshold in the *variance* invariant. Translating to an amplitude ratio requires (4); failing to do so is the easiest way to misreport a constant by a square.

9 Passive modes and the ambient system

Theorem 8 concerns a 2×2 block. We now record, as an analytic statement, how that block sits inside an n -dimensional system — and we are careful about what is proved versus what is formalized.

Proposition 17 (Determinant factorization). *Let M be symmetric and Hurwitz on $\mathbb{R}^n$ with eigenmodes $\psi_1, \dots, \psi_n$ and rates $D_1, \dots, D_n > 0$. Suppose the forcing is*

$$Q = q_b(I + (\tau - 1)ee^\top) \quad (17)$$

with e a unit vector whose modal support lies in $\Pi = \text{span}(\psi_1, \psi_2)$, i.e. $e = c\psi_1 + s\psi_2$ with $c^2 + s^2 = 1$. Then Σ block-diagonalizes as $\Sigma|_\Pi \oplus \Sigma|_{\Pi^\perp}$ and

$$\det \Sigma = \underbrace{\frac{q_b^2 \tau [1 + \mathcal{A}(w - 2)]}{4D_1 D_2}}_{\text{active block}} \cdot \prod_{j \geq 3} \frac{q_b}{2D_j} = \frac{q_b^n \tau [1 + \mathcal{A}(w - 2)]}{2^n \prod_{j=1}^n D_j}. \quad (18)$$

Proof. In the modal basis Q has entries $\tilde{Q}_{ij} = q_b \delta_{ij} + q_b(\tau - 1)\tilde{e}_i \tilde{e}_j$ with $\tilde{e} = (c, s, 0, \dots, 0)$. Hence $\tilde{Q}_{ij} = 0$ whenever $i \leq 2 < j$ or $j \leq 2 < i$, so $\tilde{Q}$ is block diagonal. By (6), $\tilde{\Sigma}_{ij} = \tilde{Q}_{ij}/(D_i + D_j)$ inherits the same block structure, and the determinant factors. The active factor is Theorem 11; each passive mode contributes $\tilde{\Sigma}_{jj} = q_b/(2D_j)$. $\square$

Remark 18 (Formalized versus documented). Proposition 17 is proved here analytically. The machine-checked theorem (Section 15) is the *explicit active block* — Theorems 4, 8, 11 and 15. The general invariant-subspace reduction of Proposition 17 is documented analytically and is *not* formalized. We are explicit about this because the distinction is easy to blur: no claim is made that Lean verifies an arbitrary $n \times n$ invariant-subspace reduction.

9.1 Effective active dimension

Proposition 17 motivates a piece of vocabulary, which we introduce with care.

The hypothesis is not that the system is two-dimensional. The ambient dimension is n , the stationary law has full support on $\mathbb{R}^n$, and all n modes are excited — the passive ones simply at the baseline intensity q_b . What is two-dimensional is the *support of the heterogeneity*: the rank-one deformation $(\tau - 1)ee^\top$ has its modal support inside Π . We therefore say the configuration has

ambient dimension	n
heterogeneous active dimension	2

and we call Π the active subspace. This is interpretation attached to a hypothesis, not a separate theorem: nothing is claimed beyond Proposition 17. In particular we do *not* say that the system “is really two-dimensional”.

The point of the vocabulary is the negative result it makes sayable. If the excitation has modal support in three or more modes, the forcing block is no longer 2×2 , the determinant is no longer a quadratic in τ , and there is no reason for a single reciprocal coordinate to suffice. Section 12.1 makes the corresponding spectral statement precise. The one-coordinate closure is a feature of effective active dimension two and should not be extrapolated.

10 A weak-coupling law

Before specializing, we record a general fact about the onset of coupling which is independent of everything above — in particular of the two-dimensional hypothesis — and which puts the graph results in context.

Proposition 19 (Weak-coupling derivative). *Let S be symmetric, $a > 0$, and let $\Sigma(\kappa)$ solve $(aI + \kappa S)\Sigma + \Sigma(aI + \kappa S) = Q$ for a fixed positive-definite Q . Then*

$$\frac{d}{d\kappa} \left[\log \det \Sigma(\kappa) \right]_{\kappa=0} = -\frac{\text{tr } S}{a}, \quad (19)$$

independently of Q .

Proof. At $\kappa = 0$ the equation gives $\Sigma_0 = Q/(2a)$. Differentiating at $\kappa = 0$, $S\Sigma_0 + \Sigma_0 S + 2a\Sigma' = 0$, so $\Sigma' = -(S\Sigma_0 + \Sigma_0 S)/(2a)$. Hence

$$\frac{d}{d\kappa} \log \det \Sigma \Big|_0 = \text{tr}(\Sigma_0^{-1} \Sigma') = -\frac{\text{tr}(\Sigma_0^{-1} S \Sigma_0) + \text{tr } S}{2a} = -\frac{2 \text{tr } S}{2a} = -\frac{\text{tr } S}{a}. \quad \square$$

For a graph Laplacian, $\text{tr } L = 2|E|$, so (19) reads $-2|E|/a$: the initial logarithmic rate of change of the generalized variance counts edges and is blind to the noise profile entirely. In particular it is strictly negative, so infinitesimal coupling always contracts the stationary covariance volume — which is the general form of the two-node statement $A'(0) < 0$ of [7], there proved for K_2 only. Any enhancement, here as there, is necessarily a finite-coupling phenomenon; this is what makes the minimization over κ in Section 11 meaningful rather than vacuous.

11 Graph corollaries

We now specialize to a concrete family. Let G be a connected graph on n nodes with Laplacian L , and take

$$M = -(aI + \kappa L), \quad a > 0, \kappa \geq 0, \quad (20)$$

so the modal rates are $D_j = a + \kappa \lambda_j$ for Laplacian eigenvalues $0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1}$. Let the forcing be (17) with e a node indicator: heterogeneity at a single node. Define

$$\Phi(\kappa) = \prod_{j \geq 1} \frac{a}{a + \kappa \lambda_j}, \quad V(\kappa, \tau) = \frac{\det \Sigma(\kappa, \tau)}{\det \Sigma(0, \tau)}. \quad (21)$$

Corollary 20 (Graph form of the threshold). *Suppose the node indicator e has modal support in exactly two eigenmodes, with rates D_1, D_2 and weights c^2, s^2 . Then*

$$V(\kappa, \tau) = \Phi(\kappa) [1 + \mathcal{A}(\kappa)(w - 2)], \quad \mathcal{A}(\kappa) = c^2 s^2 \nu(\kappa)^2, \quad (22)$$

and $V > 1$ exactly when $w > w^(\kappa) = 2 + (1 - \Phi)/(\Phi \mathcal{A})$.*

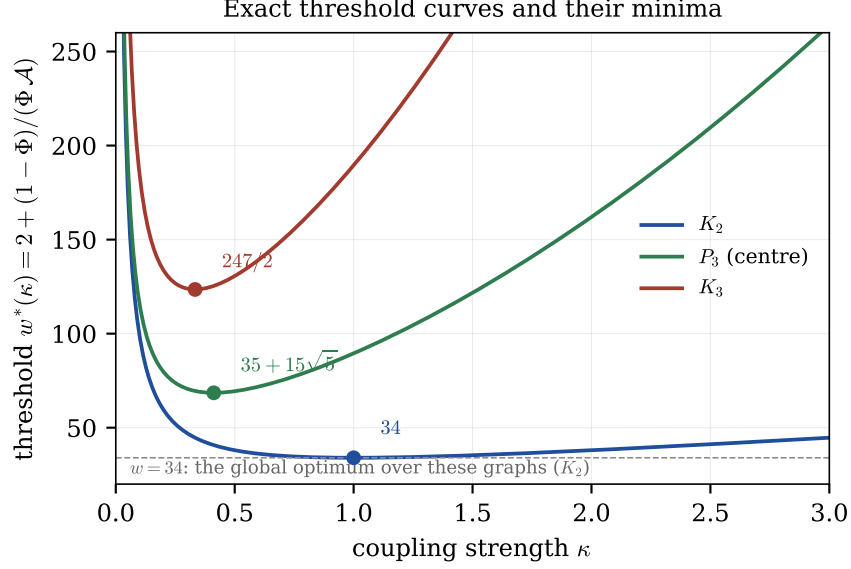


Figure 3: Exact threshold curves $w^*(\kappa)$ for K_2 , P_3 (centre node forced) and K_3 , with their exact minima marked. Each curve is the rational function derived in Sections 11.1–11.3; the marked points are the exact algebraic constants, not numerical optima.

Proof. By (18), $\det \Sigma(\kappa, \tau) = q_b^n \tau [1 + \mathcal{A}(w - 2)] / (2^n \prod_j D_j)$. At $\kappa = 0$ all rates equal a , so $\nu = 0$, $\mathcal{A} = 0$, and $\det \Sigma(0, \tau) = q_b^n \tau / (2^n a^n)$. Dividing gives (22) with $\prod_j a / D_j = \Phi$. Theorem 15 then applies. $\square$

The optimal threshold for a given graph and forced node is $w_{\min}^* = \min_{\kappa > 0} w^*(\kappa)$: the least heterogeneity that can ever produce enhancement, optimized over coupling strength. Figure 3 shows the three curves computed below.

11.1 K_2 : recovering the two-node model

For K_2 with $a = 1$, the Laplacian eigenvalues are 0 and 2, so $D_1 = 1$, $D_2 = 1 + 2\kappa$, and $\Phi = 1/(1 + 2\kappa)$. A node indicator has weights $c^2 = s^2 = \frac{1}{2}$ on the two modes, so $c^2 s^2 = \frac{1}{4}$ and

$$\mathcal{A}(\kappa) = \frac{1}{4}\nu^2, \quad \nu = \frac{1 - (1 + 2\kappa)}{1 + (1 + 2\kappa)} = \frac{-\kappa}{1 + \kappa}.$$

Corollary 20 gives

$$V(\kappa, \tau) = \frac{1}{1 + 2\kappa} \left[1 + \frac{1}{4} \frac{\kappa^2}{(1 + \kappa)^2} (w - 2) \right], \quad (23)$$

which is exactly the volume ratio of [7] after the substitution $\tau = r^2$. Minimizing $w^*(\kappa)$ over $\kappa > 0$ gives

$$\kappa^* = 1, \quad w_{\min}^* = 34. \quad (24)$$

By (4), $w = 34$ corresponds to $r + 1/r = 6$, i.e. $r = 3 + 2\sqrt{2}$, which is exactly the threshold of [7]. The coupling $\kappa^* = 1$ is the tangency at which the enhancement window of that paper closes to a point.

Remark 21. This is the sense in which the two-node result is a corollary rather than a separate theorem. It is also a check on Remark 3: the constant 34 is a value of $w = \tau + 1/\tau$, while 6 is a value of $r + 1/r$, and $34 = 6^2 - 2$.

11.2 P_3 : the path on three nodes

For P_3 the Laplacian has spectrum $\{0, 1, 3\}$ with orthonormal eigenvectors

$$\psi_0 = \frac{1}{\sqrt{3}}(1, 1, 1), \quad \psi_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad \psi_2 = \frac{1}{\sqrt{6}}(1, -2, 1).$$

Force the *centre* node, $e = (0, 1, 0)$. Its modal components are $\langle \psi_0, e \rangle = 1/\sqrt{3}$, $\langle \psi_1, e \rangle = 0$, $\langle \psi_2, e \rangle = -2/\sqrt{6}$. The vanishing middle component is the crucial structural fact: *the centre-node excitation has no support on the $\lambda = 1$ mode*, so its active support is exactly two-dimensional and Proposition 17 applies with

$$c^2 = \frac{1}{3} \ (\lambda = 0), \quad s^2 = \frac{2}{3} \ (\lambda = 3), \quad D_1 = 1, \quad D_2 = 1 + 3\kappa,$$

the $\lambda = 1$ mode being passive. Then $\nu = -3\kappa/(2 + 3\kappa)$ and

$$\mathcal{A}(\kappa) = \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{9\kappa^2}{(2 + 3\kappa)^2} = \frac{2\kappa^2}{(2 + 3\kappa)^2}, \quad \Phi(\kappa) = \frac{1}{(1 + \kappa)(1 + 3\kappa)}.$$

Substituting into $w^* = 2 + (1 - \Phi)/(\Phi\mathcal{A})$ and simplifying gives the exact curve

$$w_{P_3}^*(\kappa) = \frac{27\kappa^3 + 72\kappa^2 + 64\kappa + 16}{2\kappa}. \quad (25)$$

Corollary 22 (P_3 constant). $\min_{\kappa > 0} w_{P_3}^*(\kappa) = 35 + 15\sqrt{5}$, attained at $\kappa^* = \frac{\sqrt{5} - 1}{3}$.

Proof. The exact factorization

$$27\kappa^3 + 72\kappa^2 + 64\kappa + 16 - (35 + 15\sqrt{5}) \cdot 2\kappa = 9\left(\kappa - \frac{\sqrt{5}-1}{3}\right)^2 (3\kappa + 6 + 2\sqrt{5})$$

holds identically. For $\kappa > 0$ the second factor is positive, so the left side is nonnegative, giving $w_{P_3}^*(\kappa) \geq 35 + 15\sqrt{5}$; the double root shows equality at $\kappa^* = (\sqrt{5} - 1)/3$. $\square$

11.3 K_3 : the triangle

For K_3 the spectrum is $\{0, 3, 3\}$. A node indicator has $c^2 = 1/3$ on ψ_0 and the remaining weight $2/3$ inside the *degenerate* $\lambda = 3$ eigenspace. Because that eigenspace is two-dimensional with a single common rate, one may rotate within it so that one basis vector carries the entire projection; the excitation then has support in exactly two modes and Proposition 17 applies with $c^2 = 1/3$, $s^2 = 2/3$, $D_1 = 1$, $D_2 = 1 + 3\kappa$. The third mode is passive with the same rate $1 + 3\kappa$. Hence $\mathcal{A}$ is as for P_3 while Φ differs:

$$\mathcal{A}(\kappa) = \frac{2\kappa^2}{(2 + 3\kappa)^2}, \quad \Phi(\kappa) = \frac{1}{(1 + 3\kappa)^2},$$

giving the exact curve

$$w_{K_3}^*(\kappa) = \frac{81\kappa^3 + 162\kappa^2 + 112\kappa + 24}{2\kappa}. \quad (26)$$

Table 1: The three worked graph corollaries. All constants are exact and derived from Corollary 20; none is fitted. “Formalized” refers to the Lean development of Section 15.

Model	Active rates	c^2, s^2	κ^*	$w_{\min}^*$	Formalized
K_2	1, $1 + 2\kappa$	$\frac{1}{2}, \frac{1}{2}$	1	34	yes
P_3 (centre)	1, $1 + 3\kappa$	$\frac{1}{3}, \frac{2}{3}$	$\frac{\sqrt{5}-1}{3}$	$35 + 15\sqrt{5}$	yes
K_3 (any node)	1, $1 + 3\kappa$	$\frac{1}{3}, \frac{2}{3}$	$\frac{1}{3}$	$\frac{247}{2}$	yes

Corollary 23 (K_3 constant). $\min_{\kappa > 0} w_{K_3}^*(\kappa) = \frac{247}{2}$, attained at $\kappa^* = \frac{1}{3}$.

Proof. $81\kappa^3 + 162\kappa^2 + 112\kappa + 24 - \frac{247}{2} \cdot 2\kappa = 81\kappa^3 + 162\kappa^2 - 135\kappa + 24 = 3(3\kappa - 1)^2(3\kappa + 8)$, which is nonnegative for $\kappa > 0$ and vanishes at $\kappa = 1/3$. $\square$

Remark 24 (Degenerate eigenspaces). The K_3 computation uses a small but real subtlety. A node indicator in K_3 is not supported on two *eigenvectors* for any fixed choice of basis; it is supported on ψ_0 and on a two-dimensional eigenspace. The hypothesis of Proposition 17 is nevertheless met, because within a degenerate eigenspace any orthonormal basis is admissible and one may choose it adapted to the projection. Where the relevant eigenvalues are *distinct*, as in P_3 , no such freedom exists and the vanishing of a modal component is a genuine constraint.

11.4 Summary of the corollaries

Table 1 collects the results. Numerically $35 + 15\sqrt{5} \approx 68.54$ and $247/2 = 123.5$, both larger than the K_2 value 34: among these three configurations, the two-node system is the one for which the least heterogeneity suffices. We deliberately stop at three worked cases rather than cataloguing families of graphs; see Section 14 for what we do *not* claim about general graphs.

12 Two-dimensional spectral closure

A separate two-dimensional phenomenon explains why an earlier spectral argument closed with a single scalar. For a positive-definite 2×2 matrix with eigenvalues $\mu_1 \geq \mu_2 > 0$, write $q = \mu_1/\mu_2 \geq 1$ and

$$J_2(\Sigma) = \frac{(\text{tr } \Sigma)^2}{\det \Sigma}. \quad (27)$$

Proposition 25. $J_2 = q + \frac{1}{q} + 2$, and $q \mapsto q + 1/q$ is strictly increasing on $[1, \infty)$. Hence J_2 determines q uniquely.

Proof. $(\mu_1 + \mu_2)^2/(\mu_1\mu_2) = \mu_1/\mu_2 + \mu_2/\mu_1 + 2$. Strict monotonicity of $q + 1/q$ on $[1, \infty)$ is elementary. $\square$

This is standard. Scale-free trace/determinant functionals of a covariance matrix are the classical sphericity diagnostics, going back to Mauchly [14] and treated textbook-style by Muirhead [15, Ch. 8]; trace/determinant eigenvalue bounds of this type are likewise classical [23, 13]. We claim no novelty. It is recorded because it is the reason a single scale-free scalar suffices in dimension two — the same reason, in a different guise, that one reciprocal coordinate suffices in Theorem 11.

12.1 Failure in higher dimension

Proposition 26 (Non-identifiability for $n \geq 3$). *For $n = 3$ the pair $(\text{tr } \Sigma, \det \Sigma)$ does not determine the spectral shape: the positive triples $(1, 8, 12)$ and $(2, 3, 16)$ have the same trace 21 and the same determinant 96, hence the same value of every scale-free invariant built from trace and determinant alone, but different extreme-eigenvalue ratios 12 and 8.*

This too is background mathematics, not a new theorem. By Newton’s identities a spectrum is determined by all n elementary symmetric functions, of which trace and determinant are the complete list only when $n = 2$; the failure above is the elementary consequence. The invariant theory behind such statements is classical, and the statistical form is Mauchly’s [14, 15].

We stress the relation to Section 9.1. Proposition 26 is about the *spectrum* of an $n \times n$ covariance; Proposition 17 is about the *support of the heterogeneity*. They are different statements, but they point the same way: the one-coordinate closure used throughout this paper is a two-dimensional phenomenon. If the excitation genuinely spans three or more modes with distinct rates, neither the palindromic determinant nor the spectral closure is available, and we make no claim about that case.

13 Relation to existing work, and what we claim

Standard ingredients. The following are classical and are used as such: the stationary OU covariance and the Lyapunov equation [11, 20, 16]; the modal solution (6) for symmetric drift, standard in heterogeneous-OU work [3, 12, 9]; the determinant as Wilks’s generalized variance [22] and its entropy reading [5]; graph Laplacian eigenmodes and their use in network dynamics [2]; the reduction of a self-reciprocal polynomial through $t + 1/t$ [21]; the trace/determinant shape statistic [14, 15, 23, 13].

Closest located prior art. The nearest genuine antecedent is the work of Summers, Cortesi and Lygeros on submodularity and controllability in complex networks [17], which proves that log det of the controllability Gramian is a monotone increasing and submodular *set* function of the chosen input columns (see also [1]). That is the qualitative predecessor of the present result, and it differs in four ways that matter: it concerns a controllability Gramian rather than the stationary covariance solving (2); it is a set function over discrete input subsets rather than a continuous one-parameter family; the forcing is *added* with no baseline constraint, whereas (3) reshapes a fixed isotropic baseline; and it yields no closed form, no palindromic structure and no threshold. We are sharpening a qualitative monotonicity statement into an exact identity in a narrower setting, not contradicting or subsuming it.

Two-temperature linear Langevin systems [12, 10, 8] share the model class exactly — in particular [12, §2.10] gives the explicit 2×2 stationary covariance for arbitrary friction and diffusion — but their scalar diagnostics are entropy production, irreversibility and heat flux, not the covariance determinant. Likewise [4] analyses precisely the two-degree-of-freedom two-temperature geometry studied here and tracks an individual variance entry rather than the generalized variance. Low-rank structure in Lyapunov solutions is well studied [16, 18], but for solution approximation and singular-value bounds rather than determinants.

What we claim. Conservatively: we did not identify, in the literature we searched, an equivalent closed-form statement of the palindromic determinant identity (10), of its reciprocal reduction (12), or of the resulting exact generalized-variance threshold optimized over a coupling parameter. We describe this as an *apparently distinct assembly of standard ingredients* and make no priority

claim. Each individual ingredient is standard, and we have cited it as such. Should an equivalent theorem exist, the appropriate reading of this paper is as an exact derivation and machine-checked formalization of it, not as a new result.

Relation to the author’s other work. The two-node model of [7] is recovered in Section 11.1 as the corollary $\tau = r^2$, $c^2 = s^2 = 1/2$; that paper is the motivating special case and is cited rather than reproduced. Separately, [6] records a reciprocal invariant $\text{tr}(T)\text{tr}(T^{-1})$ determining a two-dimensional shape ratio up to inversion, and is relevant as conceptual background on one-coordinate completeness in two dimensions. We state explicitly that the present theorem is derived independently from Lyapunov covariance dynamics; [6] is invariant-theoretic context, not a source for it.

14 Limitations

1. The results are exact statements about a linear-Gaussian model. No nonlinear, biological, cognitive, social or physical-universality claim is made or implied, and no significance is attached to the numerical values of the constants beyond this model class.
2. Theorems 4–15 require an *exactly* two-dimensional active support. If the excitation has modal support in three or more modes with distinct rates, the determinant is not a quadratic in τ and nothing here applies.
3. The forcing baseline is isotropic *on the active plane*. A non-isotropic baseline breaks the self-duality of Proposition 6 and with it the palindromic structure.
4. The drift is assumed symmetric on the active plane. Non-normal drift is outside the scope of the modal formula (6).
5. Proposition 17 — the n -dimensional reduction — is proved analytically but is *not* machine-checked. Only the 2×2 active block is formalized (Remark 18).
6. The formalization verifies algebra, not stochastic-process theory. That the Σ of (2) is the stationary covariance of (1) is classical and is assumed, not verified in Lean.
7. “Effective active dimension” (Section 9.1) is interpretive vocabulary attached to a hypothesis, not an independent theorem.
8. We work out three graph corollaries. We do *not* claim a general characterization of which graphs and forced nodes satisfy the two-dimensional active-support hypothesis, nor a general lower bound on $w_{\min}^*$ over all graphs. A bound of that kind was explored in preparatory work but is not reproduced here, because we were not able to reconstruct its proof to the standard applied elsewhere in this paper; see Section A.
9. Optimizing w^* over κ treats coupling as freely tunable and the forcing geometry as fixed. Other optimizations are possible and would give different constants.

15 Machine-checked verification

The central algebraic content is formalized in Lean 4 with mathlib, in the public repository <https://github.com/innerlightr-wq/ou-threshold-lean>. Within it:

Table 2: Correspondence between the statements of this paper and the machine-checked theorems.

This paper	Lean name	Status
Theorem 4	activeCov_lyapunov, activeCov_unique	formalized
Lemma 5	det_activeForcing	formalized
Proposition 6	activeForcing_self_dual	formalized
Theorem 8	det_activeCov_palindromic	formalized
Lemma 10	two_le_reciprocalExcitation, reciprocalExcitation_eq_two_iff	formalized
Theorem 11	normalizedActiveDet_eq	formalized
Proposition 12	activeCoeff_eq_zero_iff	formalized
Proposition 13	mixing_le_quarter, rateContrast_sq_lt_one	formalized
Theorem 15	volumeRatio_gt_one_iff	formalized
Section 11.1	volume_div_volume0_eq, reciprocalExcitation_sq	formalized
Corollary 22	p3_thresholdCurve_eq, p3_threshold_min/_attained	formalized
Corollary 23	k3_thresholdCurve_eq, k3_threshold_min/_attained	formalized
Proposition 25	J2_eq_reciprocalInvariant, eigen_ratio_unique_of_J2	formalized
Proposition 26	exists_same_trace_det_different_ratio	formalized
Proposition 17	—	analytic only
Section 9.1	—	interpretation

- the results of this paper live in `OUCorridor/ActivePlane.lean` and `OUCorridor/SpectralShape.lean`;
- supporting material is in `Covariance.lean`, `Threshold.lean`, `MainTheorem.lean`, `ResidualThreshold.lean` and `CrossThreshold.lean`;
- a frozen snapshot of the exact sources, environment and build is kept at `verified/active-plane-2026-09/`.

Environment. Lean 4.33.1 (x86_64-unknown-linux-gnu), with mathlib pinned at revision

0df444a360eaa60ab8c11dca51a86af692955474.

The recorded build completes with 8715 jobs.

Audit. A grep of the library for `sorry`, `admit` and `axiom` returns no matches. All 28 audited theorems — the 22 of `ActivePlane.lean` and `SpectralShape.lean` together with the pre-existing main theorems — report exactly the three standard axioms of mathlib’s classical foundation under `#print axioms: propext, Classical.choice, Quot.sound`. There is no custom axiom and no `sorryAx`.

What is not formalized. As stated in Remark 18 and Table 2: the n -dimensional factorization of Proposition 17, the interpretive content of Section 9.1, and all stochastic-process theory. Lean checks exact real algebra on matrices. The modelling step — that (2) describes the stationary covariance of (1) — is classical and assumed.

16 Reproducibility

The manuscript directory `paper/active-plane-theorem/` is self-contained: it holds `main.tex`, its own copy of `references.bib`, the figures, the build script, the build log, and `manuscript_checks.py`. Every displayed identity in this paper is re-derived and verified symbolically by that script using exact arithmetic (SymPy), with no floating-point step in any claim: it solves the Lyapunov equation from scratch rather than checking a pre-written solution, extracts the determinant’s coefficients in τ and compares leading against constant, derives Φ and $\mathcal{A}$ for P_3 and K_3 from the graph Laplacians, and confirms each exact minimum by algebraic factorization. The script reports 55/55 checks passing.

Exact commit hashes, the branch name and file digests are recorded in the accompanying `README.md` and `SHA256SUMS.txt` at the state in which this draft was built.

Acknowledgments

AI-assisted tools were used for symbolic checking, literature organization and citation verification, code support, construction and checking of the Lean development, and editorial assistance. Claims identified in Table 2 as formalized were checked by Lean’s kernel under the pinned environment recorded in Section 15, which does not depend on the assistant’s correctness; the remaining claims were verified symbolically by the reproducibility script. The author reviewed the manuscript and takes responsibility for it. This draft has not undergone independent human peer review.

A Open items

Recorded so that the gaps are explicit rather than silent.

1. **A general graph lower bound.** Preparatory work suggested

$$w_{\min}^* \geq 2 + \frac{64|E|}{\lambda} \geq 34, \quad (28)$$

with equality only for K_2 . The inequality chain behind it is elementary and we were able to reconstruct it: it combines $c^2 s^2 \leq \frac{1}{4}$ (Proposition 13), the AM–GM bound $(2a + \kappa\lambda)^2 \geq 8a\kappa\lambda$, and $\Phi^{-1} - 1 = \prod_{j \geq 1} (1 + \kappa\lambda_j/a) - 1 \geq \kappa \operatorname{tr} L/a$, each holding pointwise in κ and hence surviving the minimization, together with $\operatorname{tr} L = 2|E|$ and $\lambda \leq \operatorname{tr} L$.

It is nevertheless excluded from the body, because stating it correctly requires four qualifications that are easy to lose and that we did not want to carry:

- (a) λ is the Laplacian eigenvalue of the *single relative eigenmode excited by e* — the one giving $D_2 = a + \kappa\lambda$. It is not the algebraic connectivity and not $\lambda_{\max}$. Read loosely, (28) becomes a different and unproved claim.
- (b) $|E| = \frac{1}{2} \operatorname{tr} L$ is an unweighted edge count; for a weighted Laplacian it must be read as total edge weight.
- (c) The quantifier must be “over connected graphs *and admissible rank-one excitations*”, where admissible means the excitation has modal support in exactly two eigenmodes. That hypothesis genuinely fails for some graphs and nodes, and without it there is no w^* to bound.

- (d) The derivation assumes the active plane *contains the common mode* ($D_1 = a$). Theorem 8 permits an active plane spanned by two nonzero- λ modes, and (28) as derived does not cover that case.

The “equality iff K_2 ” direction also needs writing out rather than asserting: for a connected graph with $n \geq 3$ the third inequality is strict for every $\kappa > 0$, whence $w_{\min}^* > 34$. Finally the bound is quantitatively weak away from K_2 (it gives 44.67 for P_3 against the true 68.54, and 66 for K_3 against 123.5), so its only real content is the corollary that 34 is a global minimum attained only by K_2 . Nothing above depends on it.

2. **Characterizing admissible configurations.** Which pairs (graph, forced node) give a node indicator with modal support in exactly two eigenmodes? The P_3 centre node works because of a symmetry-induced vanishing; K_3 works because of eigenvalue degeneracy. A general criterion is open, and is a question about graph symmetry rather than about the present theorem.
3. **Beyond rank one.** For a rank- k deformation with $k \geq 2$, or an excitation spanning three or more distinct-rate modes, the determinant is a polynomial of higher degree in the heterogeneity parameters and no single reciprocal coordinate is expected to suffice. We have not investigated whether a weaker structured reduction survives.
4. **Non-normal drift.** The modal formula (6) assumes a symmetric drift on the active plane. The non-normal case is open here.

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